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Git

Git is a version control system that tracks changes in your code, so if something breaks, you can revert to an earlier version. Managing your files and projects can be difficult. It's especially important to have a backup if you're doing a job for someone else. Git can be used to solve these problems. In team projects, Git is especially useful because it lets each person work on their own branch without affecting others. The main branch is where everything comes together.

To use Git, you first need to turn your project into a Git repository. Git has many commands. You can turn your project into a Git repository by running "git init." If you're working with an existing repository, you can use "git clone" to copy it to your IDE and start using Git's features. You can use other simple commands to manage your work. For example "git commit" saves your new changes locally. After committing your changes, you still need to push the changes to the main repository. This is helpful for teams to make changes and is used by many large companies. To combine your work with others, you'll need to use the "merge" command. However, if changes in your branch conflict with another branch, a merge conflict will occur, which needs to be resolved before merging. This might mean "stashing" some code or reverting back to earlier versions to be able to continue with the merge. This is because Git helps catch these problems early to avoid bigger issues down the line.

Git can also be used for school work. I was introduced to Git in college and have made several repositories from my projects over the years. I also used Git to collaborate with group members on a project for class and using Git simplified the process and shortened the amount of time we spent on the project. Everyone was able to work independently while still working together.

Learning Git is essential for anyone entering the computer science field. While there are advanced features, mastering the basics will take you far. It is such a good platform for companies to use to hire employees as well. Many employers ask for applicants' Github accounts to look at their range of projects. Platforms like GitHub make managing repositories easy, and once you understand how it works, the process becomes smooth.